Circuit Breaker : Multiple Choice Questions

Objective Questions > Multiple Choice Questions of Circuit breakers (1-160):

1. A circuit breaker is

(A) power factor correcting device

(B) a device to neutralize the effect of transients

(C) a waveform correcting device

(D) a current interrupting device.

D

2. The function of protective relay in a circuit breaker is

(A) to each any stray voltages

(B) to close the contacts when the actuating quantity reaches a certain predetermined value

(C) to limit arcing current during the operation of circuit breaker

(D) to provide additional safety in the operation of circuit breaker.

B

3. Low voltage circuit breakers have rated voltage of less than

(A) 220 V (B) 400V (C) 1000 V (D) 10,000 V.

C

4. The fault clearing time of a circuit breaker is usually

(A) few minutes (B) few seconds

(C) one second (D) few cycles of supply voltage.

D

5. The medium employed for extinction of arc in air circuit breaker is

(A) SF_6 (B) Oil (C) Air (D) Water.

C

6. Which of the following circuit breakers is preferred for EHT application

(A) Air blast circuit breakers (B) Minimum oil circuit breakers

(C) Bulk oil circuit breakers (D) SF_6 oil circuit breakers.

D

7 For high voltage,ac circuit breakers,the rated short circuit current is passed for

(A) 0.01 sec (B) 0.1 sec (C) 3 seconds (D) 30 seconds.

C

8. Which of the following is not a type of the contactor for circuit breakers ?

(A) Electro-magnetic (B) Electro-pneumatic

(C) Pneumatic (D) Vacuum.

D

9. Interrupting medium in a contactor may be

(A) air (B) oil (C) SF_6 gas (D) any of the above.

D

10. In air blast circuit breakers, the pressure of air is of the order of

(A) 100 mm Hg (B) 1 kg/cm^2

(C) 20 to 30 kg/cm^2 (D) 200 to 300 kg/cm^2 .

C

11. SF_6 gas is

(A) sulphur fluoride (B) sulphur difluoride

(C) sulphur hexafluorine (D) sulphur hexafluoride.

D

12. SF_6 gas

(A) is yellow in color (B) has pungent odor

(C) is highly toxic (D) is non-inflammable.

D

13. SF_6 gas

(A) is lighter than hydrogen

(B) is lighter than air

(C) has density 2-times as compared to that of air

(D) has density 5 limes as compared to that of air.

D

14. The pressure of SF_6 gas in circuit breakers is of the order of

(A) 100 mm Hg (B) 1 kg/cm^2

(C) 3 to 5 kg/cm^2 (D) 30 to 50 kg/cm^2.

C

15. While selecting a gas for circuit breaker, the property of gas that should be considered is

(A) high dielectric strength (B) non-inflammability

(C) non-toxicity (D) all of the above.

D

16. Out of the following circuit breakers, which one has the lowest voltage range ?

(A) Air-break circuit breaker (B) Tank type oil circuit breaker

(C) Air-blast circuit breaker (D) SF_6 circuit breaker.

A

17. Which of the following circuit breaker can be installed on 400 kV line

(A) Tank type oil circuit breaker (B) Miniature circuit breaker

(C) Vacuum circuit breaker (D) Air blast circuit breaker.

D

18. In a vacuum circuit breaker, the vacuum is of the order of

(A) 10mm Hg (B)10^{-2}mmHg

(C) $l0^{-6}$ mmHg (D)10^{-9}mmHg.

D

19. In modem EHV circuit breakers, the operating time between instant of receiving trip signal and final contact separation is, of the order of

(A) 0.001 sec (B) 0.015 sec (C) 0.003 sec (D) 0.03 sec.

D

20. In a HRC fuse the time between cut-off and final current zero, is known as

(A) total operating time (B) arcing time

(C) pre-arcing time (D) any of the above.

B

21. Fusing factor for a HRC fuse is

(A) Minimum fusing current / Current rating

(B) Minimum fusing current / Minimum rupturing time

(C) Maximum fusing current / Minimum fusing current

(D) Minimum fusing current / Prospective current of circuit.

A

22. Which of the following circuit breakers does not use pneumatic operating mechanism

(A) Air blast circuit breaker (B) SF_6 blast circuit breaker

(C) Air break circuit breaker (D) Bulk-oil circuit breaker.

D

23. The contact resistance of a circuit breaker is. of the order of

(A) 20 micro ohms ± 10 (B) 20milli ohms ± 10

(C) 20 ohms ± 10 (D) 200 ohms ± 10.

A

24. The insulation resistance of high voltage circuit breaker is

(A) 1k Ohm (B) 10 k Ohm (C) 20 Mega ohms (D) 2000 Mega ohm.

D

25. There is definite objection to use of which of the following medium for extinguishing the arc in case of a circuit breaker ?

(A) Air (B) SF_6 gas (C) Vacuum (D) Water.

D

26. In a circuit breaker if the insulation resistance between phase terminal and earthed frame is less than the specified limit, the probable cause could be

(A) moisture

(B) dirty insulation surface

(C) carbon or copper particles sticking to the internal surface

(D) any of the above.

D

27. If a circuit breaker does not operate on electrical compound, the probable reason could be

(A) spring defective (B) trip circuit open

(C) trip latch defective (D) any of the above.

D

28. The normal frequency rms voltage that appears across the breaker poles after final arc extinction has occurred, is

(A) recovery voltage (B) re striking voltage

(C) supply voltage (D) peak voltage.

A

29. The transient voltage that appears across the contacts at the instant of arc extinction is called

(A) recovery voltage (B) re striking voltage

(C) supply voltage (D) peak voltage.

B

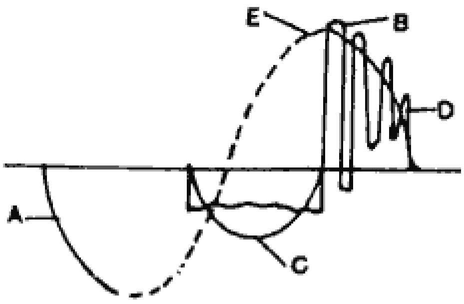
30. In a circuit breaker the active recovery voltage depends upon

(A) power factor (B) armature reaction

(C) circuit conditions (D) all of the above.

D

31. The following figure shows the voltage waveform across the pole of a circuit breaker; In this voltage R represents



(A) System voltage (B) Re striking voltage

(C) Recovery voltage (D) Extinction of arc.

B

32. In the above figure, D represents

(A) Recovery voltage (B) Re striking voltage

(C) System voltage (D) Extinction of arc.

A

33. Best protection is provided by HRC fuses in case of

(A) Open circuits (B) Short circuits

(C) Overloads (D) None of the above.

B

34. For motor circuit breakers, the time of closing the cycle is

(A) 0.001 sec (B) 0.01 sec (C) 0.10 sec (D) 0.003 sec.

D

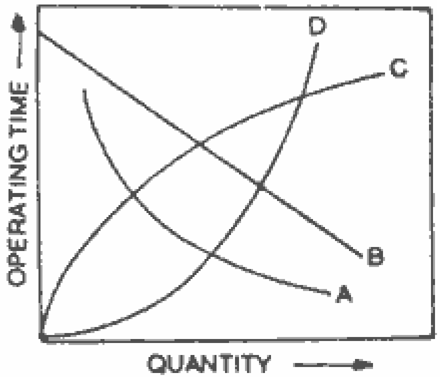
35. A relay used for protection of motors against overload is

(A) Impedance relay (B) Electromagnetic attraction type

(C) Thermal relay (D) Buchholz's relay.

C

36. Which curve in the figure represents inverse time characteristics ?



(A) Curve A (B) Curve B (C) Curve C (D) Curve D.

A

37. Fuse protection is used for current ratings up to

(A) 10 A (B) 20 A

(C) 50 A (D) 100 A.

D

38. The fuse current in amperes is related with fuse wire diameter D as

(A) I ∞ I/D (B) I ∞ D (C) I ∞ $D^{3/2}$ (D) I ∞ D^2.

C

39. The acting contacts for a circuit breakers are made of

(A) Stainless steel (B) Hard pressed carbon

(C) Porcelain (D) Copper tungsten alloy.

D

40. Ionization in a circuit breaker is not facilitated by

(A) high temperature of surrounding medium

(B) material of contacts

(C) increase of field strength

(D) increase of mean free path.

B

41. Which circuit breaker is generally used in railway traction ?

(A) SF_6 gas circuit breaker (B) Air break circuit breaker

(C) Vacuum circuit breaker (D) Minimum oil circuit breaker.

B

42. A fuse wire should have

(A) Low specific resistance and high melting point

(B) Low specific resistance and low melting point

(C) High specific resistance and high melting point

(D) High specific resistance and low melting point.

D

43. Fuse wire, protection, system is usually not used beyond

(A) 10 A (B) 25 A

(C) 50 A (D)100A.

D

44. For extra high voltage lines which circuit breaker is preferred

(A) Bulk oil circuit breaker (B) Vacuum circuit breaker

(C) SF_6 gas circuit breaker (D) Minimum oil circuit breaker.

C

45. The number of cycles in which a high speed circuit breaker can complete its operation is

(A) 3 to 8 (B) 10 to 18 (C) 20 to 30 (D) 40 to 50.

A

46. When D is the diameter of fuse wire, the fusing current will be proportional to

(A) 1/D (B) $1/D^2$ (C) $D^{3/2}$ (D) $D^{1/2}$.

C

47. A material best suited for manufacturing of fuse wire is

(A) Aluminium (B) Silver (C) Lead (D) Copper.

B

48. In a circuit breaker the current which exists at the instant of contact separation is known as

(A) re striking current (B) surge current

(C) breaking current (D) recovery current.

C

49. A Merz-price protection is suitable for

(A) transformers (B) alternators

(C) feeders (D) transmission lines.

B

50. 'Kick fuse' has

(A) square law characteristics

(B) linear characteristics

(C) inverse characteristics.

C

51. Air blast circuit breakers for 400 kV. Power system are designed to operate in

(A) 50 micro seconds (B) 50 milli seconds

(C) 500 milli seconds (D) 50 seconds.

B

52. Breaking capacity of a circuit breaker is usually expressed in terms of

(A) Amperes (B) Volts (C) MW (D) MVA.

D

53. Sulphur hexafluoride is a

(A) Conductor of electricity (B) Semi-conductor

(C) Inactive gas (D) Dielectric.

D

54. The contact resistance is least affected by

(A) the mechanical force applied

(B) the shape of the contact faces

(C) the amount of surface contamination

(D) the ambient temperature.

D

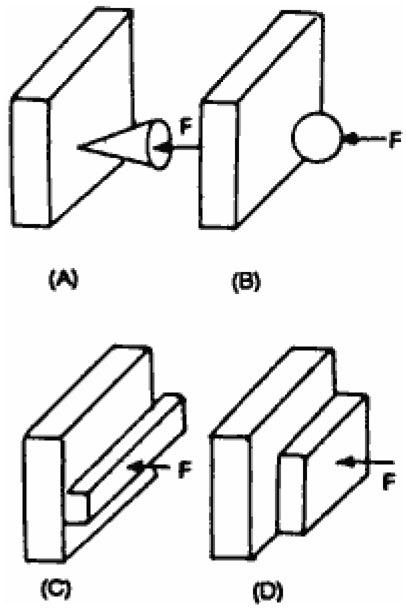
55. The arc voltage produced in ac circuit breaker is

(A) leading the arc current by 90° (B) lagging the arc current by 90°

(C) In phase with the arc current (D) In phase opposition to the arc current.

C

Questions 56 to 59 refer to the following figure :



56. Various forms of contacts are shown in the figure above. Point contact is (are) represented by

(A) A only (B) A and B only (C) A, B and C only (D) A, B, C and D.

B

57. Which contact surface provides line contact only?

(A) A (B) B (C) C (D) D.

C

58. Which form of contact are widely used in switchgear particle ?

(A) A and B only (B) A and C only (C) C and D only (D) B, C and D only.

C

59. For the various types of contacts Shown, for the same force, F, the contact resistance will be least in case of

(A) A (B) B (C) C (D) D.

A

60. As the force on contact is increased, the contact resistance will

(A) increase linearly (B) increase exponentially

(C) remain unaltered (D) decrease.

D

61. The heat produced at the contact point, due to passage of current, will least depend on

(A) contact resistance (B) time during which the current flows

(C) current flowing (D) temperature of the surrounding medium.

D

62. For the contact and their material, which of the following should have low value

(A) Contact resistance (B) Thermal capacity

(C) Thermal conductivity (D) All of the above.

A

63. Minimum arcing voltage will be least in case of

(A) carbon (B) graphite (C) tungsten (D) silver.

D

64. Minimum arcing voltage for platinum is 16 V. It can be therefore concluded that when the voltage is below 16 V

(A) it will not be possible to interrupt the circuit

(B) it will not be possible to pass the current

(C) it will be possible to interrupt any value of current without arcing

(D) it will be possible to interrupt any value of current without bringing contact closer to one another.

C

65. Oil immersion of contacts is the method of

(A) arc dispersion (B) arc prevention (C) de ionization (D) none of the above.

A

66. Which of the following is not the method of arc dispersion ?

(A) Oil immersion of contacts (B) Magnetic blow out of arc

(C) Use of rectifiers (D) De ionization of arc path.

C

67. Which of the following contact point metals has the highest melting point ?

(A) Silver (B) Tungsten (C) Gold (D) Copper.

B

68. The arc voltage produced in the circuit breaker is always

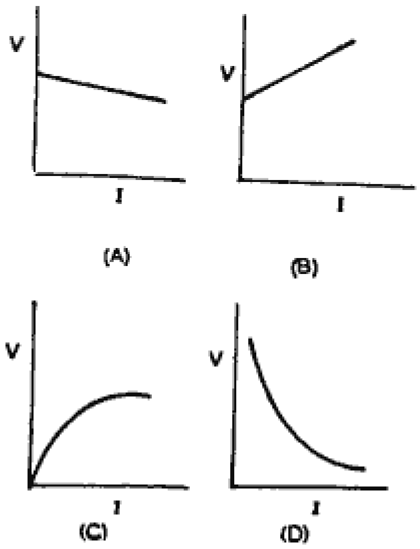
(A) in phase with arc current

(B) leading the arc current by 90°

(C) lagging the arc current by 90° .

A

69. Which of the following figure represents the voltage-current characteristics of arc in a circuit breaker ?



(A) Figure A (B) Figure B (C) Figure C (D) Figure D.

A

70. Ionization process during arc is generally accompanied by emission of

(A) light (B) heat (C) sound (D) all of the above.

D

71. Sparking between contacts can be reduced by

(A) inserting resistance in the line

(B) inserting a capacitor in series with the contacts

(C) inserting a capacitor in parallel with the contacts.

C

72. For magnetic blow out of arc the magnetic field is produced

(A) in the load circuit

(B) parallel to the axis of the arc

(C) at right angles to the axis of the arc.

C

73. Cool gases are solids brought into the arc stream assist in quenching the arc mainly by

(A) reducing current density (B) providing arc shield

(C) De ionization (D) providing parallel paths.

C

74. Sparking occurs when a load is switched off because the circuit has

(A) high inductance (B) high capacitance (C) high resistance.

A

75. HRC fuses are

(A) High resistance and Capacitance fuses

(B) Heat reflecting cool fuses

(C) Holding and Resisting current fuses

(D) High rupturing capacity fuses.

D

76. Which of the following metals does not amalgamate with mercury ?

(A) Tungsten (B) Molybdenum

(C) Nickel alloy (D) All of the above.

D

77. For the same current, which of the following fuse wires will have the least fusing time ?

(A) 18 SWG TIN - 12.5 A (B) 20 SWG TIN - 10 A

(C) 22 SWG TIN-7.5 A (D) 24 SWG TIN-5 A.

D

78. An automatic device that operates at present values is known as

(A) mercury switch (B) relay

(C) fuse (D) contactor.

B

79. The basic function of a circuit breaker is to

(A) produce the arc

(B) ionize the surrounding air

(C) transmit voltage by arcing

(D) extinguish the arc.

D

80. In a circuit breaker the arc is indicated by the process of

I. Thermal emission **II.** Ionization of oil

III. High temperature of air **IV.** Field emission

(A) I and II (B) I, II and III (C) II, III and IV (D) I and IV

D

81. The power factor of the arc in circuit breaker is

(A) always zero (B) always unity

(C) always lagging (D) always leading.

B

82. Air blast circuit breakers are usually used for

(A) instantaneous duty (B) permanent break

(C) intermittent duty (D) repeated duty.

D

83. Flame proof switch gears are usually preferred

(A) on transmission lines of low voltage (B) substations

(C) in mines (D) in high MVA capacity circuits.

C

84. Pressure of air in air blast circuit breakers is usually

(A) 1 - 5 kg/cm^2 (B) 5 - 10 kg/cm^2

(C) 10-30 kg/cm^2 (D) 35-100 kg/cm^2.

C

85. Air used in air blast circuit breaker

(A) must have least carbon dioxide (B) must be ionized

(C) must have oil mist (D) must be free from moisture.

D

86. In a circuit breaker the time duration from the instant of fault to the instant of energizing of the trip coil is known as

(A) lag time (B) lead time (C) protection time (D) operation time.

C

87. In a circuit breaker the lime duration from the instant of the fault to the extinction of arc is known as

(A) operation time (B) total clearing time

(C) lag time (D) lead time.

B

88. In a circuit breaker the lime duration from the instant of fault to the instant of closing of contact is known as

(A) Recycle time (B) Total time

(C) Gross time (D) Re closing time.

D

89. For high speed circuit breakers the total lime is nearly

(A) Half cycle (B) One cycle (C) Few cycles (D) Ten cycles.

A

90. For a high speed circuit breaker the total clearing time is nearly

(A) 1 to 2 cycles (B) 5 to 10 cycles

(C) 10 to 15 cycles (D) less than 50 cycles.

A

91. If the power factor is zero, the active recovery voltage will be

(A) minimum (B) 0.5 (C) 0.707 (D) maximum.

D

92. Which of the following is not a part of the circuit breaker ?

(A) Explosion pot (B) Fixed and moving contacts

(C) Conservator (D) Operating mechanism.

C

93. A circuit breaker will normally operate

(A) When the switch is put on (B) When the line is to be checked

(C) When the power is to be supplied (D) Whenever fault in the line occurs.

D

94. Which of the following circuit breaker will produce the least arc energy ?

(A) Minimum oil circuit breaker (B) Air blast circuit breaker

(C) Plain oil circuit breaker (D) All will produce same energy.

B

95. For a circuit breaker 'break time' is

(A) same as opening time

(B) opening time + arc duration

(C) opening time + arc duration + resistor current duration.

C

96. The breaking capacity of a circuit breaker in MVA (3 phase) is given by

(A) rated service voltage x rated symmetrical current

(B) 1.1 x rated service voltage x rated symmetrical current

(C) $(2)^{1/2}$ x rated service voltage x rated symmetrical current

(D) $(3)^{1/2}$x rated service voltage x rated symmetrical current.

D

97. Which relay is used for feeders ?

(A) MHO relay (B) Translay relay

(C) Merz price protection (D) Buchholz relay.

B

98. Which of the following relays is used on transformers ?

(A) Buchholz relay (B) MHO relay (C) Merz price relay (D) None of the above.

A

99. MHO relay is used for

(A) rectifiers (B) circuit breakers (C) transmission lines (D) feeders.

C

100. Merz-price protection is used on

(A) substations (B) capacitor bank (C) induction motor (D) generators.

D

101. Match the following :

Relay	**Operation**
(a) Static relay	(i) Responds to vector difference between two electrical quantities
(b) Over current relay	(ii) Quick operation
(c) Differential relay	(iii) Responds to increase in current
(d) Instantaneous	(iv) No moving parts relay

(A) a - (i), b- (ii), c - (iii), d - (iv)

(B) a - (iv), b - (iii), c - (i), d- (iv)

(C) a - (ii), b - (i) c- (iii), d - (iv)

(D) a - (iii), b - (ii), c - (i), d - (iv).

B

102. The values of fault current depend on

(A) voltage at the faulty point (B) total impedance up to the fault

(C) both (A) and (B) above (D) none of the above.

C

103. The advantage of neutral earthing is

(A) simplified design of earth fault protection

(B) over-voltages due to lightning can be discharged to the earth

(C) freedom from persistent arcing grounds

(D) all of the above.

D

104. Match the following :

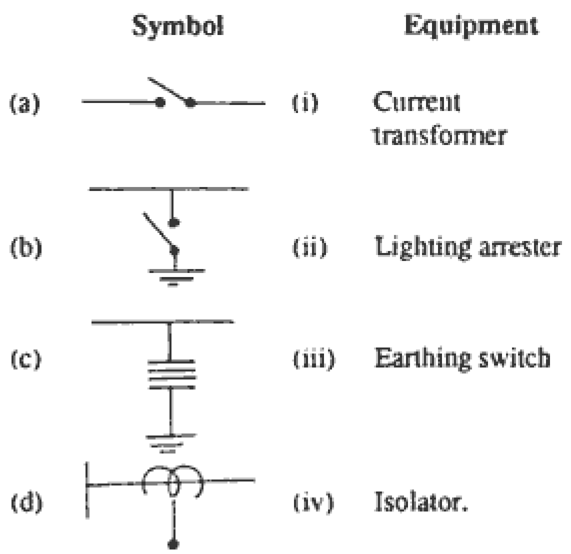
Material used in circuit breaker	Application
(a) Poly tetra	(i) Bearing surfaces fluoroethylene and sliding parts
(b) SF_6 gas	(ii) Insulating medium
(c) Electrolytic	(iii) Main contacts copper
(d) Dielectric oil	(iv) Quenching medium

(A) a- (i), b - (ii), c - (iii), d - (iv)   (B) a - (ii), b - (i), c - (iii), d - (iv)

(C) a - (iii), b - (ii), c - (i), d - (iv)   (D) a - (iv), b - (iii), c - (ii), d - (i).

A

105. Match the following:



(A) a - (i), b -(ii), c - (iii), d - (iv)

(B) a - (ii). b - (iii), c - (i), d - (iv)

(C) a - (iii), b - (i), c - (iv), d - (ii)

(D) a - (iv), b - (iii), c - (ii), d - (i).

D

106. The over-voltage surges in power systems may be caused by

(A) lightning (B) switching (C) resonance (D) any of the above.

D

107. The protection against over-voltage due to lightening is provided by

(A) use of surge diverters (B) low tower footing resistance

(C) use of overhead ground wires (D) any of the above.

D

108. Which of the following is a conducting medium for electric current ?

(A) Low temperature gas (B) High temperature gas

(C) Dissociated gas (D) Plasma.

D

109. In circuit-breakers the contact space is ionized by

(A) thermal ionization of gas

(B) thermal emission from surface of contacts

(C) field emission from the surface of contacts

(D) any of the above.

D

110. Which of the following are air-break switching devices ?

(A) Isolator (B) Limit switch (C) Earthing switch (D) All of the above.

D

111. Which of the following statement about SF_6 gas is incorrect ?

(A) It is non-toxic gas (B) It is non-inflammable

(C) It has density 5 times that of air at 20^0C (D) It has dark yellow color.

D

112. SF_6 gas is transported in

(A) gas cylinders (B) liquid form in cylinders

(C) solid form in boxes (D) air cylinders.

B

113. During arc extinction SF_6 gas

(A) decomposes into S and F ions (B) decomposes into SF_4 and SF_2

(C) gets oxidized (D) reduces to SF_3.

B

114. Dielectric strength of SF_6 is

(A) less than that of air at atmospheric pressure

(B) less than that of oil used in OCB

(C) more than that of oil used in OCB

(D) more at lower pressure and low at higher pressures.

B

115. Which of the following is the demerit of SF_6 circuit breakers ?

(A) sealing problem of gas

(B) In flux of moisture in the gas system is dangerous

(C) Deterioration of quality of circuit breaker affects reliability of circuit breaker

(D) All of the above.

D

116. Sphere gaps are used for

(A) measurement of high dc voltages (B) measurement of high ac voltages

(C) measurement of impulse voltages (D) all of the above.

D

117. Flash point of dielectric is usually above

(A) 80°C (B) 100°C (C) 140°C (D) 240°C.

C

118. A fuse is normally a

(A) current limiting device (B) voltage limiting device

(C) power limiting device (D) power factor correcting device.

A

119. Most of the fuses operate due to

(A) heating effect of current (B) magnetic effect of current

(C) electrostatic effect of current (D) none of the above.

A

120. Normally the fuse elements are in parts which are connected in the middle by ten bridge. The melting point of tin bridge is

(A) 35°C (B) 88°C (C) 230°C (D) 540°C.

C

121. The material used for bus bars should have

(A) low resistivity (B) higher softening temperature

(C) low cost (D) all of the above.

D

122. Insulation resistance of HV circuit breaker is more than

(A) 100 Ohms (B) 1 M Ohms

(C) 500 kOhms (D)100M Ohms.

D

123. The isolator is interlocked with circuit breaker and earthing switch. While opening the circuit .... opens first, then the .... and only after this the........................ can close

(A) isolator circuit breaker.................. earthing switch

(B) earthing switch .... isolator................ circuit breaker

(C) circuit breaker................. earthing switch .... isolator

(D) circuit breaker .... isolator .... earthing switch.

D

124. The main factor in favour of the use of aluminium as bus bar material is

(A) its low melting point (B) its high resistivity

(C) its low cost (D) its low density.

C

125. Over-current protection for motor is provided by

(A) cartridge fuses (B) high resistance fuses

(C) over-current relay (D) all of the above.

C

126. Fuse in motor circuit provides

(A) over current protection (B) short-circuit protection

(C) open-circuit protection (D) none of the above.

B

127. In which method of starting a motor, the starting current is the maximum ?

(A) Auto-transformer (B) Star-delta starter

(C) Stator rotor starter (D) Direct-on-line.

D

Questions 128 and 129 refer to data given below

Transformers 250 kVA, 11/0.415 kV percentage impedance 4.75%.

128. The rated current for LV side fuse should be

(A) 100 A (B) 174 A (C) 200 A (D) 348 A.

D

129. The rated current for HV side fuse should be

(A) 13.1 A (B) 23.0 A (C) 48 A (D) 55.5 A.

A

130. Which of the following are the voltage waves of magnitude higher than the desirable value ?

(A) Over-voltages (B) Surges

(C) Transients (D) All of the above.

D

131. Over-voltage transients may occur due to

(A) lightning (B) switching (C) arcing grounds (D) any of the above.

D

132. Which of the following protective devices can be used against lightning surges ?

(A) Horn gap (B) Surge diverters

(C) Lightning arresters (D) Any of the above.

D

133. Switching surges may be caused by

(A) closing of unchanged line (B) load shedding at receiving end of line

(C) switching of magnetizing current (D) any of the above.

D

134. Surge impedance of over-head transmission lines is of the order of

(A) 20 to 30 ohms (B) 300 to 500 ohms

(C) 3000 to 5000 ohms (D) 30 k ohm to 60 k ohm.

B

135. The surge impedance of under-ground cables is of the order of

(A) 20 to 60 ohms (B) 200 to 600 ohms

(C) 2 k ohm to 5 k ohm (D) 20 k ohm to 60 k ohm.

A

136. Which statement is correct

(A) SF_6 gas is nontoxic (B) SF_6 gas is lighter than air

(C) SF_6 gas has pungent smell (D) SF_6 gas is yellow in color.

A

137. The surge impedance, of a transmission line is given by

(A) $(LC)^{1/2}$ (B) $(C/L)^{1/2}$ (C) $(L/C)^{1/2}$ (D) $(L+C)^{1/2}$.

C

138. Surge modifiers are used to

(A) reduce the current of wave-front (B) reduce the voltage of wave-front

(C) reduce the steepness of wave-front (D) modify the shape of wave-front.

C

139. The steepness of the wave-front can be reduced by

(A) connecting a capacitance between line and earth

(B) connecting an inductor in series with the line

(C) either of (A) or (B) above

(D) connecting an inductor between line and earth and connecting a capacitor in series with the line.

C

140. In the circuit breaker, the arcing contacts are made of

(A) electrolytic copper

(B) copper tungsten alloy

(C) aluminium alloy

(D) porcelain.

B

141. The disadvantage offered by ungrounded systems is

(A) frequent arcing grounds

(B) difficult earth fault relaying

(C) voltage oscillations

(D) all of the above.

D

142. Solid grounding is used for voltages

(A) above 220 kV (B) above 11 kV (C) below 660 V (D) below 115 V.

C

143. Resistance grounding is used for voltages

(A) below 220 V

(B) up to 660 V

(C) between 3.3 kV to 11 kV

(D) above 66 kV.

C

144. Switching over-voltages are more hazardous than lightning surges in case of

(A) Low voltage systems

(B) 11 kV systems

(C) Unbalanced systems

(D) EHV and UHV systems.

D

145. Current limiting reactors may be

(A) air cooled, air cored

(B) oil immersed magnetically shielded

(C) oil immersed non-magnetically shielded

(D) any of the above.

D

146. Series reactors are installed at strategic locations of power systems to

(A) bring down the fault level within the capacity of switchgear

(B) directly pass the fault surges to ground

(C) pass neutralizing surges of opposite nature

(D) discharge the capacitors.

A

147. Fault diverters

(A) divert the current to earth in the event of short-circuits

(B) neutralize the surges by resistors

(C) modify the surge wave shapes

(D) none of the above.

A

148. In star connected system without neutral grounding, zero sequence currents are

(A) same as peak value of phase current

(B) same as rms value of phase currents

(C) vector sum of phase currents

(D) zero.

D

149. In which portion ofthe transmission system faults occur most frequently?

(A) Transformers (B) Overhead lines (C) Alternators (D) Underground cables.

B

150. Bulk-oil circuit breaker is suitable for voltages up to

(A)10kV (B) 16 kV (C) 26 kV (D) 36 kV.

D

151. The ohmic value of impedance to be connected in the neutral to ground circuit of a 2000 kVA transformer with earth fault relay set to 40%, with respect to 400 V side will be

(A) 0.2 ohm (B) 2.0 ohms (C) 20 ohms (D) 200 ohms.

A

Questions 152 and 153 refer to data given below:

A 3-phase, 5000 kVA, 6.6 kV generator having 12% sub-transient reactance. A 3-phase short-circuit occurs at its terminals.

152. Fault MVA is

(A) 21.5 (B) 41.66 (C) 53.33 (D) 75.75.

B

153. Fault current is

(A) 3640 A (B) 2460 A (C) frequency (D) 880 A.

A

154. The actuating quantity for the relays can be

(A) magnitude (B) phase angle (C) frequency (D) any of the above.

D

155. Electro-magnetic relays may be operated by

(A) electro-magnetic attraction (B) electro-magnetic induction

(C) thermal effect (D) any of the above.

D

156. Which of the following is not a relay using electromagnetic force

(A) Buchholz relay (B) Induction cup relay

(C) Balanced beam relay (D) Attracted armature type relay.

A

157. Buchholz relay is operated by

(A) Eddy currents (B) Gas pressure

(C) Electro-magnetic induction (D) Electro-static induction.

B

158. Thermal relays are often used in

(A) generator protection (B) transformer protection

(C) motor starters (D) none of the above.

C

159. A bimetal strip consists of two metal strips have different

(A) thermal diffusivity (B) thermal conductivity

(C) specific beat (D) coefficient of expansion.

D

160. Differential protection principle is used in the protection of

(A) generators (B) transformers (C) feeders (D) all of the above.

D

Circuit Breakers : True - False Questions

State whether the following statements related to circuit breakers are true (T) or false (F)……(1 - 40):

1. The voltage across the circuit breaker pole after final current zero is recovery voltage.

T

2. Air circuit breakers are used for voltages above 12 kV.

F

3. Double pressure type SF_6 circuit breaker is also known as Puffer type.

F

4. Air blast circuit breakers and SF_6 circuit breakers are preferred for EHV applications.

T

5. Arcing contacts of circuit-breakers are generally made of copper tungsten alloy.

T

6. The are plays important role in the behavior of circuit breaker.

T

7. The arc has a brightly burning core of high temperature ranging from 6000 K to 25000 K.

T

8. SF_6 is a heavy, colorless and chemically inert gas.

T

9. SF_6 gas is electro-positive.

F

10. In oil circuit breakers transformer oil is used.

T

11. . An isolator does not have any current making or current breaking capacity.

T

12. SF_6 circuit breaker takes minimum time for installation.

T

13. A triple pole isolator has three identical poles, with operate simultaneously.

T

14. Isolators operate under no current conditions.

T

15. Non-linear resistors are used in lightning arresters.

T

16. Switching over-voltages are proportional to system voltages.

T

17. Switching over-voltages arise from energizing of uncharged line can be reduced by the use of high voltage shunt reactors.

T

18. The damage caused by surge depends on steepness of the wave-front.

T

19. Bulk oil circuit breaker is used for outdoor UHV application.

F

20. In electro-magnetic relays the restraining torque is given by springs.

T

21. For remote operation circuit-must breaker be equipped with shunt trip.

T

22. A protective system which responds to vector difference between two electrical quantities is known as differential protection.

T

23. Protective systems prevent faults by disconnecting an equipment in the event of abnormal condition.

T

24. Selectivity is the property by virtue of which the protective relaying system distinguishes between normal condition and abnormal condition.

T

25. Frequency relays are used in generator protection.

T

26. SF_6 circuit breaker has high reliability and negligible maintenance.

T

27. Tripping relays are slow and generally attracting armature type.

F

28. The over-current relays are connected to the system normally by means of CT's.

T

29. Core balance CT's are used for earth-fault protection.

T

30. Short circuit tests are conducted on circuit breakers to prove the ratings of the circuit-breakers.

T

31. Fusing factor is the ratio of minimum fusing current to the current rating.

T

32. The distance protection responds to the ratio V/I.

T

33. Graded time lag and grade current over-current protection is used for single radial feeder where time lag can be permitted.

T

34. Distance relaying is based on measurement of impedance between relay location and fault point.

T

35. Distance relay is used where time lag can be permitted.

F

36. Pilot wire differential relaying is used for lines up to 40 km of length.

T

37. In motor starter, thermal relay provides overload protection as well as single phasing protection.

T

38. Protection against unbalanced supply voltage is provided by negative phase sequence relays.

T

39. Breaking current of a pole on a circuit breaker is the current in that pole at the instant of contact separation.

T

40. Most of the alternators are provided with Buchholz relay in addition to differential protection.

F